FEMMETABOTWIN

Women's glucose health, seen through a personal lens.

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A cycle-aware Digital Twin concept for understanding how individual glucose patterns may interact with meals, activity and menstrual-cycle context.





PERSONALISED • CYCLE-AWARE • EVIDENCE-AWARE

Prototype presentation | October 2026

Women are not just smaller versions of men

When research overlooks sex-specific biology—or fails to analyse it—important patterns can remain invisible.

31.4%

average representation of women across the nine diabetes medicine-development studies included in one systematic review

Striking for a single review—not a universal estimate for all diabetes research.

The gap is not simply who gets enrolled.

Representation

Who is included in studies?

Analysis

Are outcomes examined by sex and relevant life stage?

FemMetaboTwin's premise

Make relevant context visible, then test whether it improves personal understanding—without assuming every woman responds the same way.

Source: systematic review of diabetes-medicine development studies (2023), PubMed PMID 37413814.

Hormonal shifts can change the metabolic context

Oestrogen and progesterone vary across the cycle; these shifts may influence insulin sensitivity and glucose patterns, but effects differ between individuals.

A simplified menstrual-cycle timeline

MENSTRUAL
Cycle begins

FOLLICULAR
Oestrogen rises

OVULATORY
Hormones shift

LUTEAL
Progesterone higher

Why this may matter

Hormonal changes can influence insulin action, glucose utilization, appetite, sleep and activity—factors that may interact with daily glucose response.

Why personalization matters

Cycle effects are not universal; symptoms, cycle timing, medication, food, stress and individual physiology all matter.

Conceptual physiology, not a deterministic phase-to-glucose rule. Source: Gamarra & Trimboli, 2023 systematic review.

A signal worth studying—not a settled rule

Research in women with type 1 diabetes suggests some experience higher glucose or reduced insulin sensitivity in parts of the cycle; findings remain heterogeneous.

What has been observed

Some studies report greater hyperglycaemia or reduced insulin sensitivity during the luteal phase in a subset of women.

What remains uncertain

Small samples, differing phase definitions and mixed methods limit strong conclusions.

The important gap

Evidence in Type 2 diabetes and cycle-aware personalized prediction is not established by those T1D studies.

Therefore: learn each person’s pattern; do not hard-code the same cycle correction for everyone.

Source: Systematic review of 14 studies in T1D; authors found no conclusive evidence and highlighted high heterogeneity / risk of bias (2023).

FemMetaboTwin: a living, personal metabolic picture

A Digital Twin that evolves with the person’s observations and treats menstrual-cycle context as a potentially relevant signal—not a universal explanation.

OBSERVE

Glucose • meals • activity

CONTEXTUALISE

Cycle timing • symptoms, if available

UPDATE TWIN

Personal baseline • variability • response history

PREDICT / FLAG

Only when evidence and inputs support it

REFLECT

Compare outcome with expectation

Personal baseline

The person’s own history is the reference—not a population average alone.

Cycle-aware only when supported

If cycle data are missing or unreliable, do not apply a phase-specific adjustment.

Sex-related biology meets real-world context

Diabetes risk, presentation and management can be shaped by a mix of biology and lived context; the goal is not to reduce women to hormones.

Biology across life stages

Reproductive hormones, pregnancy, postpartum changes and menopause can alter metabolic context and care needs.

Patterns can be individual

The same cycle phase will not affect every person’s glucose in the same way; repeated personal observations matter.

Care must avoid assumptions

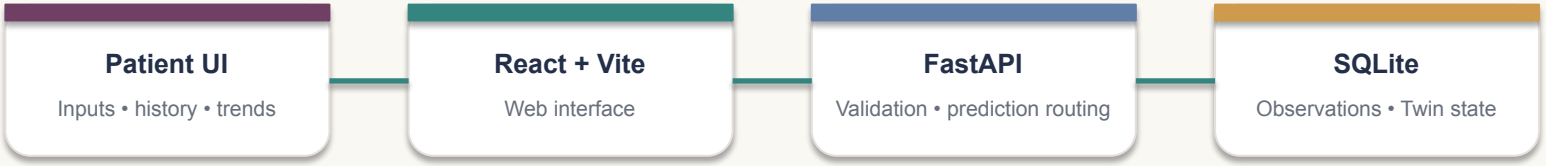
Cycle phase is one context signal alongside meals, activity, sleep, stress, medication and baseline physiology.

Design principle: represent relevant female physiology without assuming that all women menstruate or that every menstruating person has the same response.

This is a design rationale, not a claim that all sex differences are proven or that cycle effects are universal.

A practical web prototype for a future evidence-aware twin

The current application gives the concept a working interface and persistent state.



Prediction eligibility

Checks provenance, expected features and held-out evaluation evidence before serving a trained prediction.

Fallback / unavailable

Personal-mean estimate only when eligible observations exist; otherwise report unavailable.

Audit and What-If

Shows status and limitations; What-If matching does not claim causality.

Cycle-aware correction is a future evidence-gated layer, not a validated feature in the current build.

From logging to learning—not from phase to prescription

The prototype supports observation history, Twin state, forecasts/statuses and exploratory historical matching.

1 Log

Enter glucose observations and meal-carbohydrate context; records persist through the API.

2 Review

Explore recent history, trends and the current Twin state.

3 Compare

What-If uses comparable historical cases when available; otherwise it reports insufficient matches.

No medication changes. No causal claims. No hidden confidence score.

The current personal-mean fallback is not a trained ML model; uncertainty is not estimated.

Personalization must earn the right to predict

The prototype is designed to show its evidence status and degrade gracefully when evidence is missing.

- Trained predictions require provenance, feature compatibility and held-out evaluation evidence.
- Missing inputs or inference failures cannot silently become trained predictions.
- Fallbacks are labeled; if no suitable observations exist, the result is unavailable.
- Uncertainty is explicitly “Not estimated.”
- Synthetic demo scenarios are labeled and do not count as clinical evidence.
- Cycle context must be reliable before it can influence an individual prediction.

THE PROMISE

Not “women are all the same.”

Not “the cycle always changes glucose.”

A model that can learn when context matters—and admit when it cannot.

The contribution is a framework—not a clinical claim

FemMetaboTwin turns a women's-health research question into an inspectable prototype and a testable roadmap.

Built

React/Vite interface, FastAPI backend, SQLite persistence, patient history/trends, What-If matching and audit/status paths.

Verified engineering

Latest reported checks: 18 backend tests passed; frontend production build passed.

Not yet validated

No verified CGMacros observations, no real-data-trained model, no held-out metrics and no clinical validation.

NEXT RESEARCH MILESTONES

- Acquire a verified dataset and inspect the actual schema.
- Establish a simple glucose-trajectory baseline with participant-aware, time-safe evaluation.
- Study whether cycle context adds predictive value beyond meals, activity and personal baseline.
- Validate in a suitable cohort before any clinical claims.

North star: women’s glucose patterns deserve to be studied as individual, longitudinal signals—not assumed away.

Evidence behind the problem framing

Selected sources used to ground the motivation and avoid overstating the evidence.

- 1. Gender representation in drug development studies for diabetes mellitus: a systematic review (2023). PubMed PMID: 37413814.
- 2. Gamarra E, Trimboli P. Menstrual Cycle, Glucose Control and Insulin Sensitivity in Type 1 Diabetes: A Systematic Review. J Pers Med. 2023;13(2):374. doi:10.3390/jpm13020374.
- 3. National Academies of Sciences, Engineering, and Medicine. Improving Representation in Clinical Trials and Research (2022), chapter on representation in clinical research.
- 4. NIH policy on inclusion of women and members of minority groups in NIH-funded clinical research; policy and reporting guidance.
- 5. PhysioNet CGMacros, v1.0.0. Dataset and license: CC BY-NC-SA 4.0. The archive has not been verified or used to train this prototype.

Interpretation note: evidence about menstrual-cycle effects is stronger in some Type 1 diabetes studies than in Type 2 diabetes; the effect is heterogeneous and not universal.